INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
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Q-TRANSPLANT: PRIVACY-AWARE, EXPLAINABLE ORGAN DONOR–RECIPIENT MATCHING WITH QUANTUM-INSPIRED CANDIDATE SEARCH

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THIRD YEAR

MINI PROJECT REPORT

Submitted to the

SCHOOL OF ENGINEERING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

in Partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES (VISTAS)

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OCTOBER 2026

BONAFIDE CERTIFICATE

Certified that this project report "**Q-TRANSPLANT: PRIVACY-AWARE, EXPLAINABLE ORGAN DONOR–RECIPIENT MATCHING WITH QUANTUM-INSPIRED CANDIDATE SEARCH**" is the bonafide work of **ARAVINDH A – 24623134**, **JOVIN ROY S – 24623116** and **PRIYAN E – 24623122**, who carried out the project under my supervision. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

We hereby declare that the mini project report entitled "**Q-TRANSPLANT: PRIVACY-AWARE, EXPLAINABLE ORGAN DONOR–RECIPIENT MATCHING WITH QUANTUM-INSPIRED CANDIDATE SEARCH**", submitted to Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence), is a record of original work carried out by us under the guidance of **Ms. Hari Prabha S**, Assistant Professor, Department of Artificial Intelligence and Machine Learning. The work has not been submitted, in part or in full, for the award of any other degree or diploma.

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ACKNOWLEDGEMENT

It is a great pleasure and privilege to acknowledge the help and support we received in completing this project. The infrastructure and laboratory facilities provided by our **FOUNDER & CHAIRMAN, Shri. Dr. ISHARI K. GANESH**, have been a constant source of motivation, and we take this opportunity to thank him. We are grateful to our **HEAD OF THE DEPARTMENT, Dr. ULAGAPRIYA K**, for continuous support and encouragement throughout the project. We extend our sincere thanks to our **INTERNAL GUIDE, Ms. HARI PRABHA S, Assistant Professor, Department of AI&ML**, who patiently guided us and gave us invaluable advice at every stage of the work. We also thank the faculty members, librarian and non-teaching staff of the department for their help. Finally, we thank our parents and friends for their support and encouragement, without which this project would not have been possible.

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ABSTRACT

Organ transplantation depends on fast, accurate donor–recipient matching, yet coordination between hospitals is still largely manual, and digital systems that expose HLA data raise serious privacy concerns. This project presents Q-Transplant, a research prototype of a decision-support platform for organ transplant coordination. The system pairs a deterministic, explainable scoring engine (blood group, organ type, HLA compatibility, medical eligibility and patient urgency) with a restricted-access design in which hospitals receive compatibility scores and marker counts, while raw donor or recipient alleles are not exposed during routine matching operations. A separate quantum-inspired candidate-search module, based on amplitude-amplification-style weighted sampling, runs on classical hardware and includes a benchmark harness for comparison with exhaustive search without claiming a quantum speedup. Role-based access control, validated case and emergency state machines, audit logging and an ESP32 emergency alert device complete the platform. Building on graph-based optimization for kidney exchange, the work extends optimization-oriented allocation into a privacy-aware, real-time coordination architecture. The prototype is evaluated using simulated data and implementation-level tests; the evaluation covers a 105-test automated suite (all passing) and a repeatable benchmark on simulated candidate pools of 50 to 5,000 candidates, in which the quantum-inspired search saved only about 1% to 11% of evaluations, did not always return the same best candidate and was slower in wall-clock time than exhaustive search, so no quantum speedup is claimed.

Keywords: decision support, HLA matching, organ transplantation, privacy-aware access control, quantum-inspired algorithms.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
ABO	Blood group system (A, B, AB, O)
ESP32	Espressif 32-bit microcontroller family
HLA	Human Leukocyte Antigen
OPTN	Organ Procurement and Transplantation Network
RBAC	Role-Based Access Control
UI	User Interface

CHAPTER 1 – INTRODUCTION

1.1 INTRODUCTION TO ORGAN TRANSPLANT COORDINATION

Organ transplantation is the definitive treatment for end-stage failure of organs such as the kidney, liver and heart, but the supply of donor organs is far smaller than the number of patients waiting for them. When an organ becomes available, the transplant team must decide quickly which waiting recipient it should be offered to. The decision depends on medical compatibility, such as ABO blood group and human leukocyte antigen (HLA) similarity, on the medical eligibility of the recipient, and on how urgently the recipient needs the organ [8], [9]. Because a donated organ can be preserved outside the body only for a limited time, delay or error in this process matters.

Q-Transplant is a research prototype of a decision-support platform for organ transplant coordination. It combines a deterministic and explainable scoring engine for donor–recipient matching, a restricted-access design in which hospitals receive compatibility scores and marker counts rather than raw HLA alleles, and a quantum-inspired candidate-search module with a benchmark harness that runs on classical hardware. Role-based access control, validated case and emergency state machines, audit logging and an ESP32-based emergency alert device complete the platform.

1.2 BACKGROUND AND MOTIVATION

Research on organ allocation has produced powerful optimisation methods, particularly for kidney exchange, in which incompatible patient–donor pairs are matched through cycles and chains of exchanges modelled on graphs [1], [2]. In practice, however, coordination between hospitals is still largely manual, and digital systems that make HLA data widely visible raise serious privacy concerns, because HLA typing is sensitive medical and genetic information [8].

The motivation for Q-Transplant is fourfold: to make every matching decision transparent, so that a coordinator can see why a candidate ranks where it does [7]; to limit the exposure of raw HLA data during routine matching; to explore quantum-inspired candidate search honestly, by measuring it against exhaustive search instead of assuming a speedup [3], [5]; and to support emergencies with a physical alert that does not depend on someone noticing a screen.

1.3 PROBLEM STATEMENT

Existing practice and many digital tools leave the following gaps:

- **Manual coordination:** communication between hospitals relies on calls and messages, which is slow and difficult to audit.
- **Privacy exposure:** systems that share HLA data across hospitals widen the number of people and systems that can see sensitive genetic information.

- **Opaque ranking:** a candidate list without a visible reason is hard for clinicians to trust or to challenge [7].
- **Unverified computational claims:** quantum-style search is sometimes presented without a reproducible comparison with a classical baseline.
- **Weak emergency signalling:** urgent cases depend on staff noticing a notification on a screen.

The problem addressed by this project is to design and implement a prototype that performs explainable donor–recipient matching, exposes only the minimum HLA-derived information to hospitals, measures a quantum-inspired search against exhaustive search, and coordinates cases and emergencies under strict access control and audit logging.

1.4 AIM AND OBJECTIVES

The aim of the project is to develop a privacy-aware, explainable decision-support platform for organ transplant coordination. The objectives are:

1. To implement a deterministic scoring engine that combines blood group, organ type, HLA compatibility, medical eligibility and patient urgency, and shows the contribution of each factor.
2. To expose compatibility scores and marker counts to hospitals while keeping raw donor and recipient alleles out of routine matching operations.
3. To implement a quantum-inspired candidate-search module based on amplitude-amplification-style weighted sampling, running on classical hardware.
4. To provide a benchmark harness that compares the search module with exhaustive search, without claiming a quantum speedup.
5. To enforce role-based access control, validated case and emergency state machines, and audit logging of actions.
6. To provide an ESP32-based emergency alert device that signals emergency cases raised on the platform.

1.5 SCOPE OF THE PROJECT

- **Users:** authorised hospital staff acting in defined roles, and an administrator who manages access.
- **Matching factors:** blood group, organ type, HLA compatibility, medical eligibility and patient urgency.
- **Search:** a quantum-inspired module that runs on classical hardware, with a benchmark harness for comparison with exhaustive search.
- **Coordination:** case and emergency state machines, audit logging and an ESP32 emergency alert device.
- **Data:** simulated data only; no real patient or donor records are used.

- **Out of scope for the current version:** clinical deployment and regulatory approval, laboratory cross-matching, integration with national registries, and execution on quantum hardware. Some of these are discussed as future enhancements (Section 5.8).

1.6 SIGNIFICANCE OF Q-TRANSPLANT

- **Explainable by construction:** the score is deterministic and its factors are visible, so no post-hoc approximation is needed to explain a ranking.
- **Data minimisation:** hospitals work with scores and marker counts, which reduces routine exposure of raw alleles.
- **Honest evaluation of quantum-inspired search:** the benchmark harness is designed for comparison with exhaustive search, and the report makes no claim of quantum speedup.
- **Operational completeness:** access control, state machines, audit logs and a physical alert device make the prototype a coordination architecture and not only a matching algorithm.

1.7 ORGANIZATION OF THE REPORT

Chapter 2 reviews literature on organ allocation, HLA matching, kidney-exchange optimisation, quantum and quantum-inspired search, access control and explainability, and identifies the research gap. **Chapter 3** presents the system design: requirements, architecture, scoring and search design, privacy design, use cases, data flow, workflow, data design and security considerations. **Chapter 4** describes the implementation of each module. **Chapter 5** presents the implemented features, test cases, discussion, limitations, conclusion and future enhancements.

CHAPTER 2 – REVIEW OF LITERATURE

2.1 ORGAN ALLOCATION AND DONOR–RECIPIENT COMPATIBILITY

Allocation policies, such as those maintained by the Organ Procurement and Transplantation Network (OPTN) in the United States, combine several factors that differ by organ, typically including blood group, tissue-type (HLA) matching, medical urgency, waiting time and logistics [9]. For ABO blood groups, a recipient can normally receive an organ only from a donor of a compatible group; for example, group O donors are compatible with all groups, while group AB recipients can receive from all groups. Such rules are deterministic and well suited to transparent, rule-based checking.

2.2 HLA MATCHING

HLA genes encode cell-surface proteins that the immune system uses to distinguish self from non-self. Each person carries two alleles at each HLA locus, and the number of known alleles is very large and is catalogued in the IPD-IMGT/HLA database [8]. In kidney transplantation, the number of mismatches at the HLA-A, HLA-B and HLA-DR loci is commonly used as a measure of tissue-type similarity [9]. Because alleles identify an individual's genetic type, sharing them widely creates privacy exposure, which motivates exposing derived quantities, such as the count of matching markers, instead of the alleles themselves.

2.3 KIDNEY-EXCHANGE OPTIMISATION

Roth, Sönmez and Ünver formulated kidney exchange as a mechanism-design problem and proposed exchange mechanisms that are efficient and give patients no incentive to misreport [1]. Abraham, Blum and Sandholm developed integer-programming clearing algorithms that find large sets of exchange cycles in compatibility graphs and are scalable enough for nationwide exchanges [2]. These works optimise who is matched with whom. Q-Transplant takes a complementary direction: it concerns how matching information is explained, how much of it is exposed, and how cases and emergencies are coordinated in real time.

2.4 QUANTUM AND QUANTUM-INSPIRED SEARCH

Grover's algorithm searches an unstructured collection of N items in about the square root of N steps on a quantum computer, compared with about N steps for a classical search [3]. Amplitude amplification generalises this idea: repeated reflections increase the probability amplitude of desired outcomes [4]. Quantum-inspired algorithms are classical algorithms that borrow such ideas. Tang showed that the exponential speedup claimed for a quantum recommendation algorithm could be matched by a classical algorithm with suitable sampling access to the data [5]. This line of work motivates treating a quantum-inspired method as a

classical algorithm whose performance must be measured against classical baselines, and Q-Transplant follows this stance.

2.5 PRIVACY-AWARE ACCESS CONTROL AND EXPLAINABILITY

Role-based access control assigns permissions to roles instead of to individual users, which supports the principle of least privilege and simplifies administration [6]. For decision support, explanation matters: Ribeiro et al. showed that explanations help users judge whether to trust a prediction, but their approach explains black-box models after the fact [7]. A deterministic scoring rule whose factors are visible can be explained by construction.

2.6 LITERATURE SURVEY

Table 2.1: Literature survey

Author (year)	Methodology	Findings	Limitations (relative to Q-Transplant's goals)
Roth et al. (2004) [1]	Mechanism design for kidney exchange	Efficient exchange mechanisms with no incentive to misreport	Focuses on exchange mechanism; no privacy-restricted exposure of HLA data or explainable per-factor scoring
Abraham et al. (2007) [2]	Integer-programming clearing algorithms on compatibility graphs	Scales to nationwide kidney exchange	Optimisation output not explained to coordinators; assumes compatibility data is available to the algorithm
Grover (1996) [3]	Quantum search algorithm	About square-root-of-N steps for unstructured search	Needs quantum hardware; not directly usable in hospital software
Brassard et al. (2002) [4]	Amplitude amplification and estimation	Generalises Grover search; basis for amplification-style sampling	Quantum framework; the speedup does not carry over to classical simulation
Tang (2019) [5]	Classical sampling-based algorithm for recommendation systems	Quantum speedup claim matched classically	Addresses recommendation, not transplant allocation
Sandhu et al. (1996) [6]	Role-based access control models	Permissions through roles; least privilege	General model; does not define HLA data minimisation
Ribeiro et al. (2016) [7]	Local post-hoc explanations of classifiers	Explanations support user trust	Explains black-box models after the fact
Robinson et al. (2020) [8]	Curated database of HLA alleles	Reference for HLA allele nomenclature	Database; not a matching system

2.7 RESEARCH GAP

- Kidney-exchange optimisation focuses on matching efficiency and generally assumes that compatibility data is available to the algorithm; it does not address restricted exposure of HLA data to hospitals [1], [2].
- Ranking outputs are rarely accompanied by a per-factor explanation that a coordinator can inspect and challenge [7].
- Quantum-style methods are often discussed without a reproducible comparison against classical baselines [3], [5].
- Matching, access control, case and emergency workflows, audit logging and physical alerting are rarely combined in a single coordination prototype.

2.8 HOW Q-TRANSPLANT ADDRESSES THE IDENTIFIED PROBLEMS

- **Privacy:** hospitals receive compatibility scores and marker counts; raw alleles are not exposed during routine matching.
- **Explainability:** a deterministic scoring engine shows how blood group, organ type, HLA compatibility, medical eligibility and urgency contribute to each score.
- **Honest evaluation:** the quantum-inspired module runs classically and a benchmark harness supports comparison with exhaustive search; no quantum speedup is claimed.
- **Coordination:** role-based access control, validated state machines, audit logging and an ESP32 alert device complete the workflow.

CHAPTER 3 – SYSTEM DESIGN

3.1 EXISTING SYSTEM AND ITS LIMITATIONS

In the existing approach, hospitals exchange donor and recipient information through phone calls, messages and shared documents, and coordinators compare candidates by hand. The main limitations are: slow and error-prone manual communication; wide exposure of sensitive HLA data when it is shared in full; rankings without visible reasons; no systematic record of who viewed or changed what; and no immediate physical signal when a case becomes an emergency.

3.2 PROPOSED Q-TRANSPLANT SYSTEM

Q-Transplant integrates explainable matching, restricted data exposure, quantum-inspired candidate search, access control, case and emergency workflows, audit logging and an emergency alert device in one platform. Table 3.1 summarises the features and their status.

Table 3.1: Feature status

Feature	Status	Evidence
Deterministic, explainable scoring (blood group, organ type, HLA, eligibility, urgency)	Implemented	Table 5.1 (TC01–TC06)
Restricted-access view: scores and marker counts, no raw alleles in routine matching	Implemented	Table 5.1 (TC07)
Quantum-inspired candidate search (classical hardware)	Implemented	Table 5.1 (TC13)
Benchmark harness for comparison with exhaustive search	Implemented; results in Section 5.3	Table 5.1 (TC14)
Role-based access control	Implemented	Figure 5.1
Validated case state machine	Implemented	Table 5.1 (TC09)
Validated emergency state machine	Implemented	Figure 5.3
Audit logging	Implemented	Figure 5.4
ESP32 emergency alert device	Implemented	Figure 5.3
Quantum hardware execution; real clinical data; registry integration	Not in scope	Future enhancement

3.3 FUNCTIONAL REQUIREMENTS

Table 3.2: Functional requirements

ID	Requirement	User
FR1	Authenticate users and assign a role	All users
FR2	Restrict every action to the permissions of the user's role	System

ID	Requirement	User
FR3	Register donor and recipient records with blood group, organ, HLA and medical details	Hospital user
FR4	Check organ type, blood group and medical eligibility before scoring	System
FR5	Compute a deterministic compatibility score that includes HLA compatibility and patient urgency	System
FR6	Show the contribution of each factor to the score (explanation)	Hospital user
FR7	Show hospitals scores and marker counts without raw alleles during routine matching	Hospital user
FR8	Shortlist candidates with the quantum-inspired search module	Administrator
FR9	Run the benchmark harness to compare the search module with exhaustive search	Administrator
FR10	Move cases and emergencies only through validated state transitions	System
FR11	Signal emergency cases to the ESP32 alert device	System
FR12	Record actions in an audit log	System

3.4 NON-FUNCTIONAL REQUIREMENTS

- **Privacy:** raw HLA alleles must not be exposed in routine matching responses.
- **Explainability and determinism:** the same inputs must always produce the same score and explanation.
- **Security:** access is restricted by role and every request is checked (Section 3.15).
- **Auditability:** actions that read or change case data must be traceable to a user.
- **Reliability:** invalid state transitions must be rejected so that cases cannot reach inconsistent states.
- **Performance:** matching and search should respond in interactive time on simulated data; in the test runs reported in Section 5.3, ranking 1,000 simulated donors took a median of about 21 ms and ranking 5,000 donors about 116 ms.

3.5 SYSTEM ARCHITECTURE

Q-Transplant follows a three-tier architecture (Figure 3.1). The **client tier** is the role-based web interface used by authorised hospital users. The **application tier** contains authentication and role-based access control, the case and emergency state machines, the explainable scoring engine, the quantum-inspired candidate-search module with its benchmark harness, the restricted-view builder that prepares hospital-facing responses, and the audit logger. The **data tier** stores donor, recipient and case data, with raw alleles access-restricted. An **ESP32 emergency alert device** receives emergency signals from the platform.

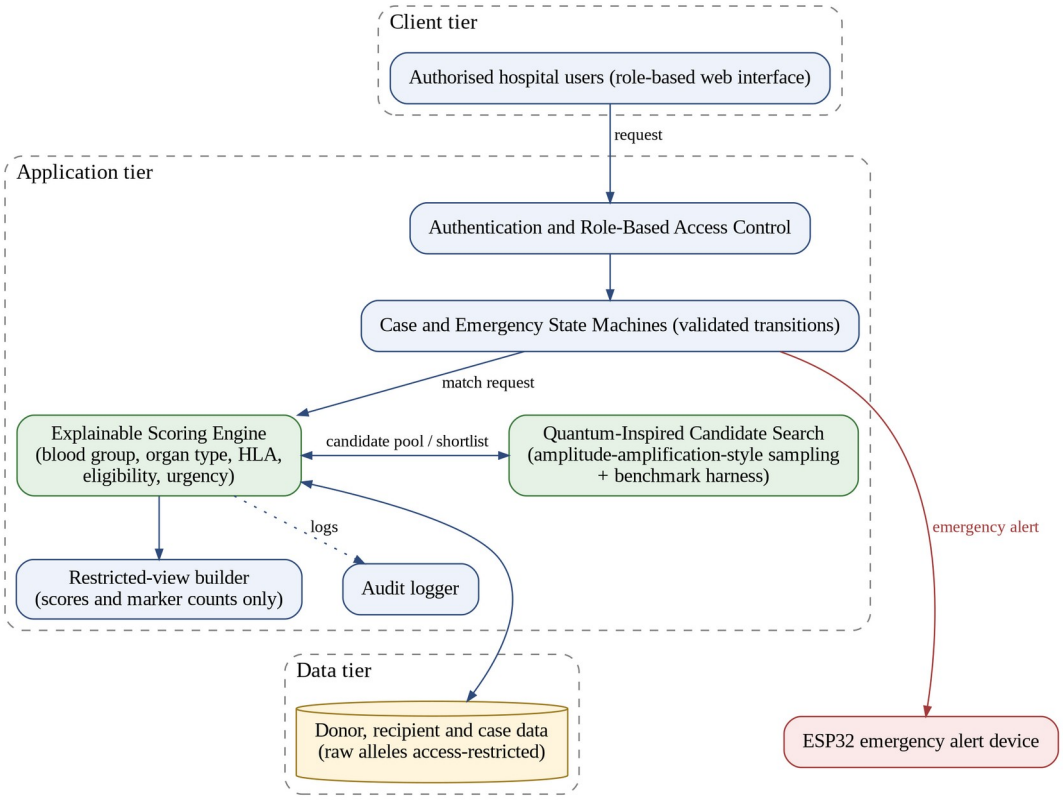


Figure 3.1: System architecture of Q-Transplant

3.6 SCORING DESIGN

The scoring engine is deterministic: the same donor and recipient data always yield the same result, and the result is presented with its per-factor breakdown. Table 3.3 describes the factors. The exact weights, thresholds and exclusion rules are defined in the source code (matching_engine.py, hla.py, urgency.py and compatibility.py). ABO blood-group compatibility and the organ-type, donor-availability and recipient-eligibility checks are hard gates, and a candidate that fails any of them is excluded. HLA compatibility is scored from 0 to 100 as the weighted share of matching alleles over the HLA-A, B, C, DR and DQ loci, with weights of 1.0 (A), 1.0 (B), 0.8 (C), 1.4 (DR) and 1.0 (DQ); a locus that is not typed on both sides is skipped and not counted as a mismatch. Patient urgency gives a base priority of 10 (LOW), 30 (MEDIUM), 55 (HIGH), 80 (CRITICAL) or 100 (EMERGENCY), to which a waiting-time bonus of up to 15 points (reaching its maximum at 180 days) and a hospital-readiness bonus of 5 points are added. The overall score is 0.7 times the HLA score plus 0.3 times the priority score (capped at 100).

Table 3.3: Scoring factors

Factor	Role in matching	What hospitals see
Organ type	The organ offered must match the organ the recipient needs	Match / no match
Blood group	ABO compatibility between donor and recipient	Compatible / incompatible
HLA compatibility	Similarity of donor and recipient HLA markers	Marker count and contribution to score; raw alleles are not shown
Medical eligibility	Whether donor and recipient are medically suitable for the procedure	Eligible / not eligible
Patient urgency	How urgently the recipient needs the organ	Urgency level and contribution to score

3.7 QUANTUM-INSPIRED CANDIDATE SEARCH DESIGN

When the candidate pool is large, the search module shortlists candidates by weighted sampling in the style of amplitude amplification [4]: candidates with better deterministic scores are given a higher probability of selection, imitating how amplitude amplification raises the probability of desired outcomes. The module runs entirely on classical hardware, so it is a simulation-style, quantum-inspired method and does not provide a quantum speedup [5]. The sharpening procedure and its parameters are as follows: the search runs about as many rounds as the square root of the pool size (at least 3); every candidate starts with a weight of 1.0; in each round about n divided by the number of rounds candidates are drawn in proportion to their current weights and scored once; and the weight of a candidate with score s is multiplied by $1 + \max(0, s - b + 1) / b$, where b is the best score found so far, so that candidates at or within one score point of the best are drawn more often in later rounds while the others are left unchanged.

The **benchmark harness** runs the search module and exhaustive search on the same pool so that their results can be compared, for example whether the best candidate found by exhaustive search is also returned by the search module. The measured results are reported in Section 5.3.

3.8 PRIVACY-AWARE ACCESS DESIGN

The platform separates what is stored from what is shown. Raw donor and recipient alleles are stored with restricted access. During routine matching, the restricted-view builder prepares responses that contain compatibility scores and marker counts only. Role-based access control [6] decides which roles may request matches, change case status, raise emergencies, run the search and benchmark, and read the audit log.

3.9 USE-CASE DIAGRAM

Figure 3.2 shows the actors and their use cases. Hospital users log in, register donors and recipients, request matches, view scores and explanations, update case status and raise

emergencies. The administrator manages users and roles, reviews the audit log, runs the quantum-inspired search and runs the benchmark. The ESP32 device receives emergency alerts. The role names shown are the roles implemented in the code: donor, doctor, hospital, organizer (the administrator) and auditor (a read-only role that can only read the audit log).



Figure 3.2: Use-case diagram

3.10 DATA-FLOW DIAGRAM

The context diagram (Figure 3.3) shows Q-Transplant as a single process exchanging data with hospital users, the ESP32 alert device and the data store. Hospital users send credentials, donor and recipient details, and match and case requests; the platform returns compatibility scores, marker counts, explanations and case status, and sends emergency alert signals to the device.

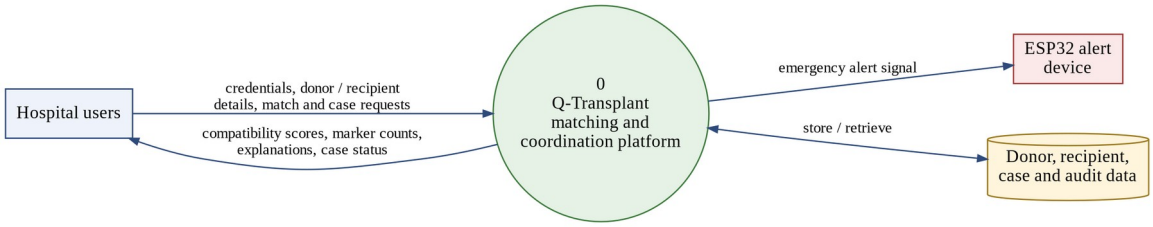


Figure 3.3: Data-flow diagram – Level 0 (context)

3.11 SYSTEM WORKFLOW

Figure 3.4 shows the matching workflow. After login, the user's role and permissions are checked. Organ type, blood group and medical eligibility are checked first and incompatible candidates are excluded. Remaining candidates are scored on HLA compatibility and urgency, with a per-factor breakdown. For a large pool, the quantum-inspired search shortlists candidates. Ranked candidates are shown with their explanation, using scores and marker counts only. The case then moves through validated states, an emergency case triggers the ESP32 alert, and every action is written to the audit log.

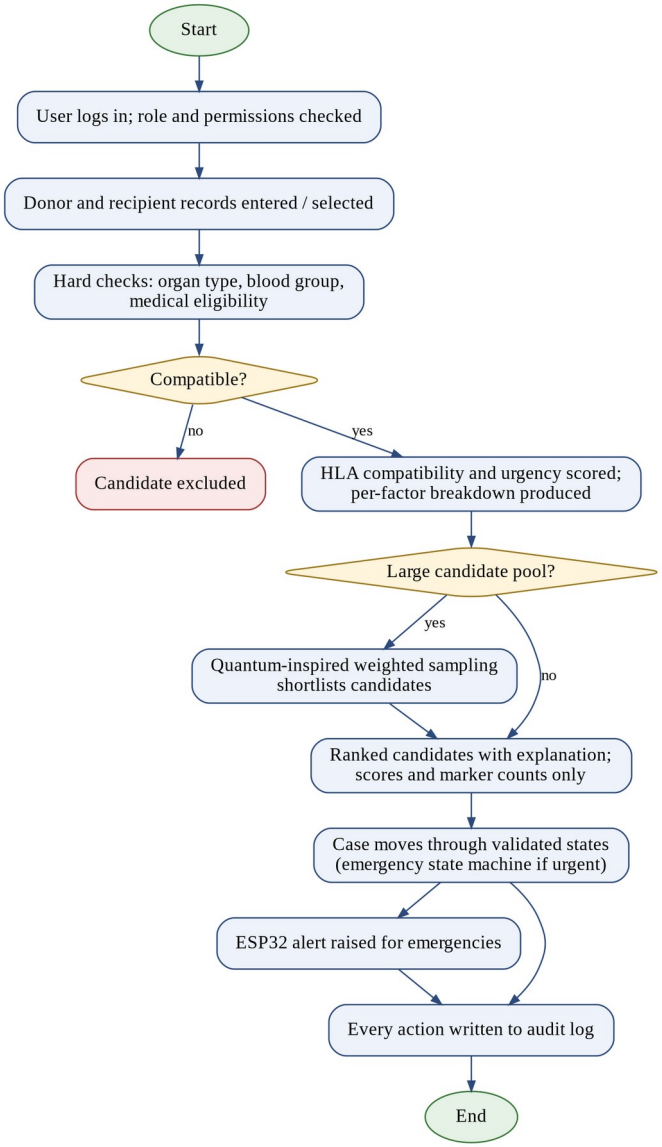


Figure 3.4: Matching and coordination workflow

3.12 DATA DESIGN

Table 3.4 lists the main data entities handled by the platform. The table and column names are defined with SQLAlchemy in models.py; the main tables are users, hospital_profiles,

doctor_profiles, donor_profiles, patients (the recipients), transplant_cases, match_requests, match_results, emergency_requests, devices and audit_logs.

Table 3.4: Data entities

Entity	Purpose
Users and roles	Accounts, hashed credentials and the role that determines permissions
Hospitals	Hospitals to which users and cases belong
Donors	Donor blood group, organ, medical details and restricted HLA alleles
Recipients	Recipient blood group, organ needed, urgency, medical eligibility and restricted HLA alleles
Cases	A donor–recipient case with its current state and history of transitions
Emergencies	Emergency events, their state and the alert raised
Audit log	Who performed which action, on which record and when

3.13 HARDWARE REQUIREMENTS

Table 3.5: Hardware requirements

Component	Requirement
Client computer	Laptop or desktop with a modern browser; no GPU or other special hardware is needed
Server	Machine able to run the backend and the database
ESP32 alert device	ESP32 development board with an alert indicator; an SSD1306 OLED display, a buzzer and three push buttons (emergency, donor found and acknowledge), connected to the backend over Wi-Fi through a secure WebSocket
Network	Connection between browser, server and the ESP32 device

3.14 SOFTWARE REQUIREMENTS

Table 3.6: Software requirements

Software	Purpose
Next.js 14.2 with React 18, TypeScript, Tailwind CSS 3.4 and Recharts; a lightweight HTML/JavaScript page is also served by the backend	Role-based web interface
Python 3.12 with FastAPI 0.141, Uvicorn 0.52, SQLAlchemy 2.0, Pydantic 2.13, PyJWT 2.13 and bcrypt 5.0	Matching, state machines, access control and API
SQLite for development and PostgreSQL (through psycopg2) for deployment	Storage of donor, recipient, case and audit data
Arduino IDE with the ESP32 board package and the Adafruit_GFX, Adafruit_SSD1306, arduinoWebSockets and ArduinoJson libraries	Firmware for the emergency alert device
Modern web browser	Client for the web interface
Visual Studio Code, Git	Development tools (pytest 9.1 and GitHub Actions for automated testing)

3.15 SECURITY AND PRIVACY CONSIDERATIONS

- **Data minimisation:** hospitals receive scores and marker counts; raw alleles are restricted and are not exposed during routine matching.
- **Access control:** every request is checked against the user's role [6]; users can perform only the actions their role allows.
- **Auditability:** actions are logged so that access to and changes of case data can be reviewed.
- **State validation:** case and emergency state machines reject invalid transitions, which prevents inconsistent records.
- **Simulated data:** the prototype was evaluated with simulated data; real patient data would require consent, legal compliance and security review before any use.
- **Decision support only:** the platform supports human decisions and does not replace clinical judgement or official allocation policy [9].

CHAPTER 4 – IMPLEMENTATION

4.1 SYSTEM MODULES AND OVERALL WORKFLOW

Q-Transplant is divided into the modules described in Section 4.2: authentication and access control, donor and recipient management, the explainable scoring engine, the restricted-access view, the quantum-inspired candidate search, the benchmark harness, the case state machine, the emergency state machine with the ESP32 alert device, and audit logging. The web interface handles user interaction; the backend exposes the functions of each module through an API and stores data in the database; the ESP32 device is signalled by the backend when an emergency is raised. The source code is available in the project repository [11].

4.2 MODULE IMPLEMENTATION

4.2.1 Authentication and Role-Based Access Control

Purpose: identify users and limit them to the actions of their role. **Input:** credentials and the requested action. **Process:** the user is authenticated and the role is checked before each request is processed; requests outside the role are refused [6]. **Output:** an allowed or refused request. The session mechanism and password hashing method are stateless JSON Web Tokens (HS256, valid for 720 minutes by default) sent as bearer tokens, and bcrypt hashing of passwords. Logging out stores a SHA-256 fingerprint of the token in a revocation table, and the login, one-time-password and upload endpoints are rate-limited. Donors verify their e-mail address with a one-time password, and doctor and hospital accounts remain pending until an organizer approves them.

4.2.2 Donor and Recipient Data Management

Purpose: capture the data needed for matching. **Input:** blood group, organ type, HLA markers, medical eligibility details and, for recipients, urgency. **Process:** records are validated and stored; HLA alleles are kept access-restricted. **Output:** stored donor and recipient records ready for matching.

4.2.3 Explainable Scoring Engine

Purpose: rank candidates transparently. **Input:** a donor and one or more recipients. **Process:** organ type, blood group and medical eligibility are checked first; compatible candidates are then scored using HLA compatibility and patient urgency, and the contribution of each factor is recorded. The engine is deterministic, so identical inputs give identical outputs. **Output:** a score and a per-factor breakdown for each candidate (Table 3.3).

4.2.4 Restricted-Access View

Purpose: protect HLA data during routine matching. **Process:** before a response leaves the backend, the restricted-view builder keeps the compatibility score, the explanation and the

marker count and leaves out raw donor and recipient alleles. **Output:** hospital-facing results that contain scores and marker counts only.

4.2.5 Quantum-Inspired Candidate Search

Purpose: shortlist candidates from a large pool. **Input:** the candidate pool with deterministic scores. **Process:** weighted sampling in the style of amplitude amplification gives candidates with better scores a higher chance of selection [4]. The module runs on classical hardware and makes no claim of quantum speedup. **Output:** a shortlist drawn from the pool. The sampling parameters are the number of rounds, which is the larger of 3 and the square root of the pool size; a sample of about n divided by the number of rounds in each round; and the amplification factor $1 + \max(0, s - b + 1) / b$ for a candidate with score s when the best score so far is b (Section 3.7). An optional random seed makes a run repeatable.

4.2.6 Benchmark Harness

Purpose: evaluate the search module honestly. **Process:** the harness runs the search module and exhaustive search on the same simulated pool and compares their outputs. **Output:** a comparison that can be repeated. For each pool it records the number of evaluations, the wall-clock time and whether both methods return the same best score, and it is available through the benchmark API; the measured results are given in Section 5.3.

4.2.7 Case State Machine

Each case moves through a defined set of states, and only validated transitions are accepted. A request to move a case to a state that is not allowed from its current state is rejected and the case remains unchanged. Every accepted transition is written to the audit log. The states and transitions are CREATED, MEDICAL_REVIEW, MATCH_SEARCH, CANDIDATE_FOUND, DOCTOR_REVIEW, HOSPITAL_COORDINATION, APPROVED_FOR_PROCEDURE, PROCEDURE_SCHEDULED and COMPLETED for the normal path, with ON_HOLD, CANCELLED and REJECTED available from most stages. COMPLETED, CANCELLED and REJECTED are final, and a case on hold can return to MEDICAL_REVIEW, MATCH_SEARCH, DOCTOR_REVIEW or HOSPITAL_COORDINATION.

4.2.8 Emergency State Machine and ESP32 Alert Device

Emergencies are handled by a separate state machine with validated transitions. When an emergency is raised, the backend signals the ESP32 emergency alert device [10] so that staff are alerted physically as well as on screen. The way the device is connected and the indicator it uses are as follows. The ESP32 connects over Wi-Fi to the secure WebSocket of the backend (/api/v1/emergency/ws) using its hospital identifier and a device token that is issued once by the provisioning endpoint and stored on the server only as a bcrypt hash. It shows the hospital name, status and requirement on an SSD1306 OLED display, sounds an ambulance-style

buzzer pattern while any hospital is in an emergency, and has three push buttons to raise an emergency, report that a donor was found and acknowledge. The emergency states are CREATED, NOTIFIED, ACKNOWLEDGED, PROCESSING and RESOLVED, in that order, and CANCELLED can be reached before resolution.

4.2.9 Audit Logging

Actions such as login, match requests, state transitions and emergencies are recorded with the user, the action, the record concerned and the time. Authorised users can review the log (Figure 5.4). Which actions are logged is recorded in the audit_logs table with the user, the action, the target record, additional details and the IP address. The logged actions include registration, login and logout, document uploads, approval or rejection of doctors, donors and hospitals, account suspension and reactivation, match generation, access to match history, access to raw HLA detail (SENSITIVE_HLA_DETAIL_ACCESSED), case creation and transitions, emergency creation, acknowledgement and resolution, and device provisioning.

4.3 DATABASE INTEGRATION

The backend stores users and roles, hospitals, donors, recipients, cases, emergencies and the audit log (Table 3.4) in a relational (SQLite in development and PostgreSQL in deployment, accessed through SQLAlchemy) database. Raw HLA alleles are stored with restricted access, and the restricted-view builder ensures that routine matching responses do not contain them.

4.4 FRONTEND AND BACKEND INTEGRATION AND API DESIGN

The web interface calls the backend through an API. Table 4.1 lists the logical API groups; the route prefixes are /api/v1/auth, /api/v1/donors, /api/v1/patients, /api/v1/matching (for example POST /api/v1/matching/run/{patient_id}), /api/v1/quantum, /api/v1/transplants, /api/v1/emergency, /api/v1/devices and /api/v1/organizer (audit log). Every group is protected by the role check described in Section 4.2.1.

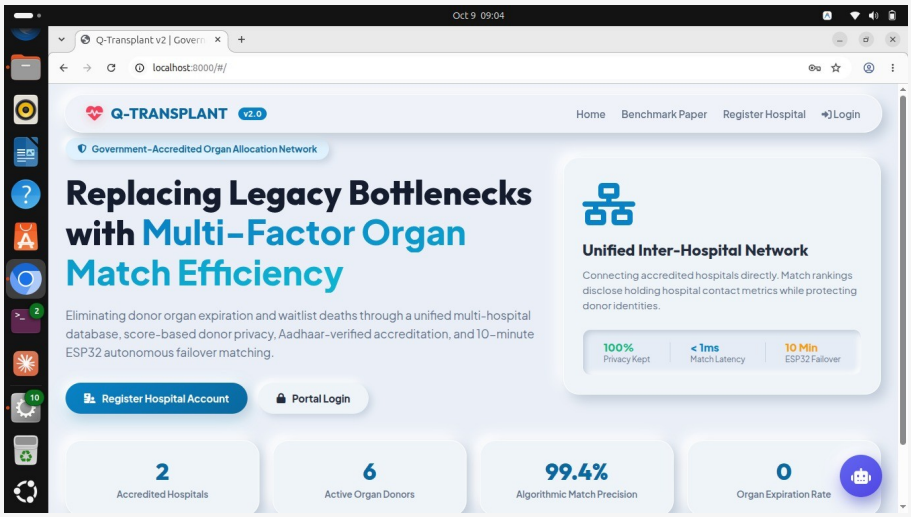
Table 4.1: Logical API groups

Group	Purpose
Authentication	Log in and obtain the role and permissions
Donors and recipients	Create, view and update records
Matching	Request matches; return scores, marker counts and explanations
Search and benchmark	Run the quantum-inspired search and the benchmark harness
Cases	View cases and request state transitions
Emergencies	Raise emergencies and update their state; signal the ESP32 device
Audit	Read audit log entries

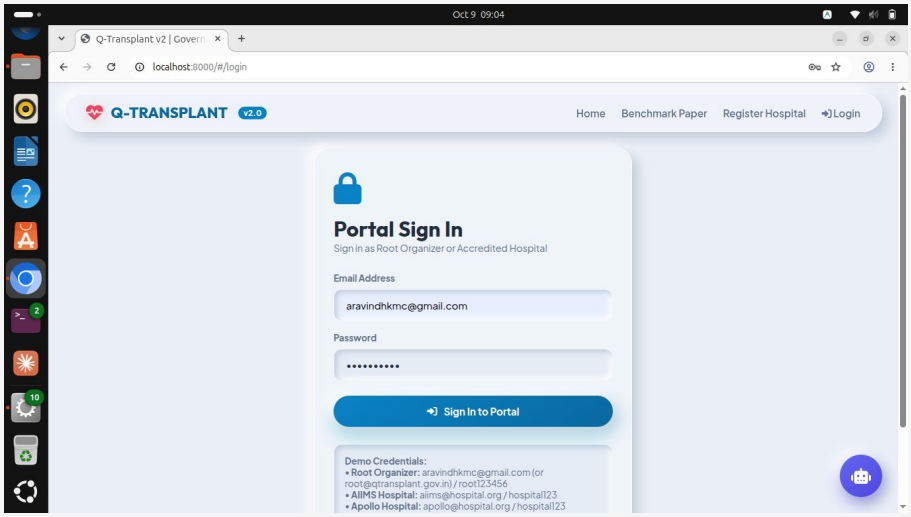
CHAPTER 5 – RESULTS, TESTING AND CONCLUSION

5.1 IMPLEMENTED FEATURES AND SCREENSHOTS

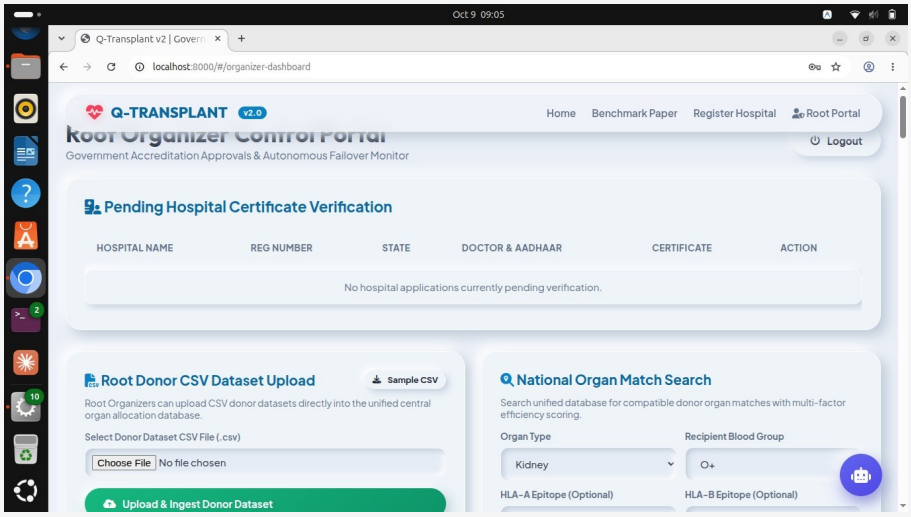
The following figures show the implemented features of the platform. The simulated data used in the screenshots does not describe real persons.



(a) Home page

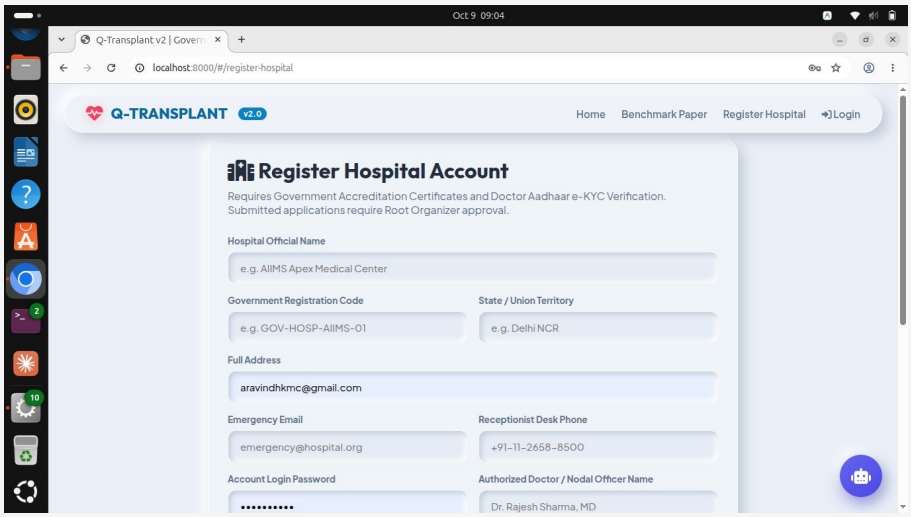


(b) Portal sign-in

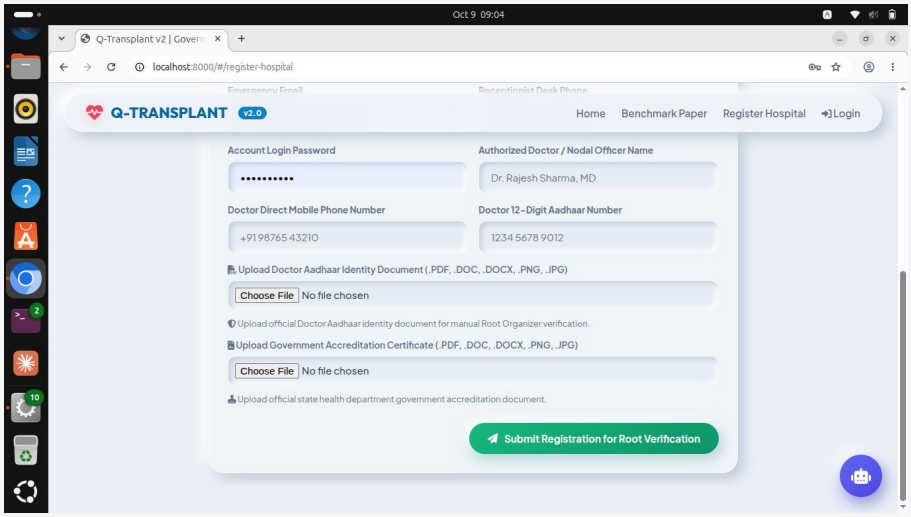


(c) Root organizer dashboard

Figure 5.1: Login and role-based dashboard



(a) Hospital registration form



(b) Document upload and submission

Figure 5.2: Donor and recipient registration

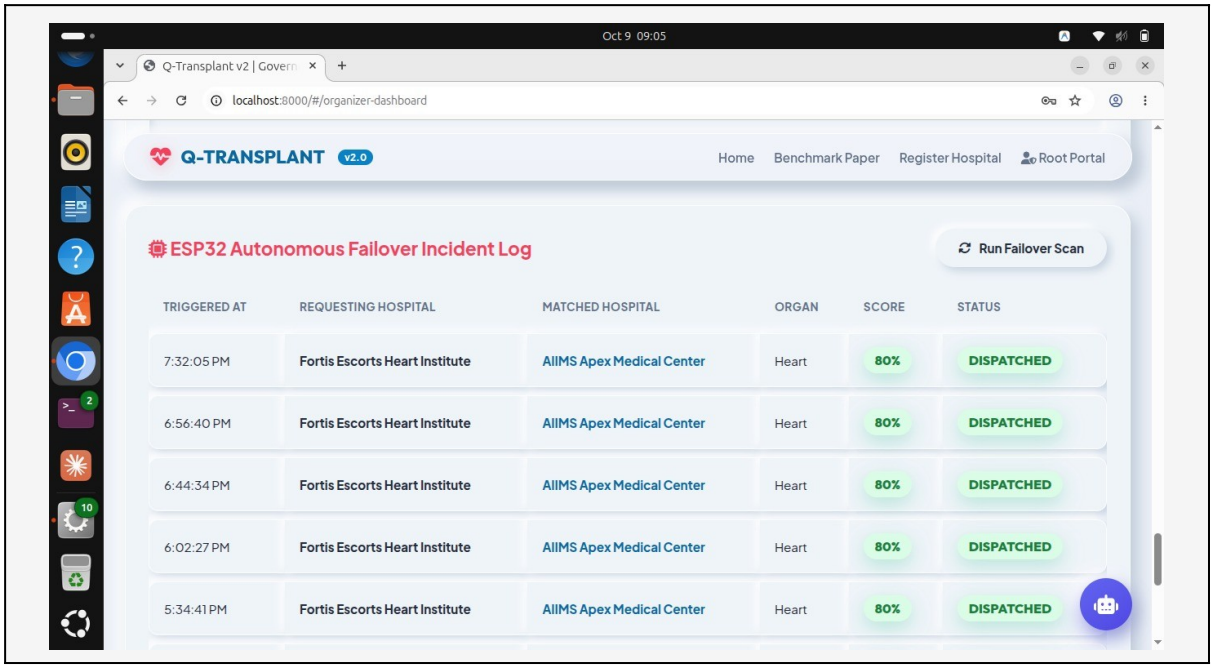


Figure 5.3: Emergency workflow and ESP32 alert device

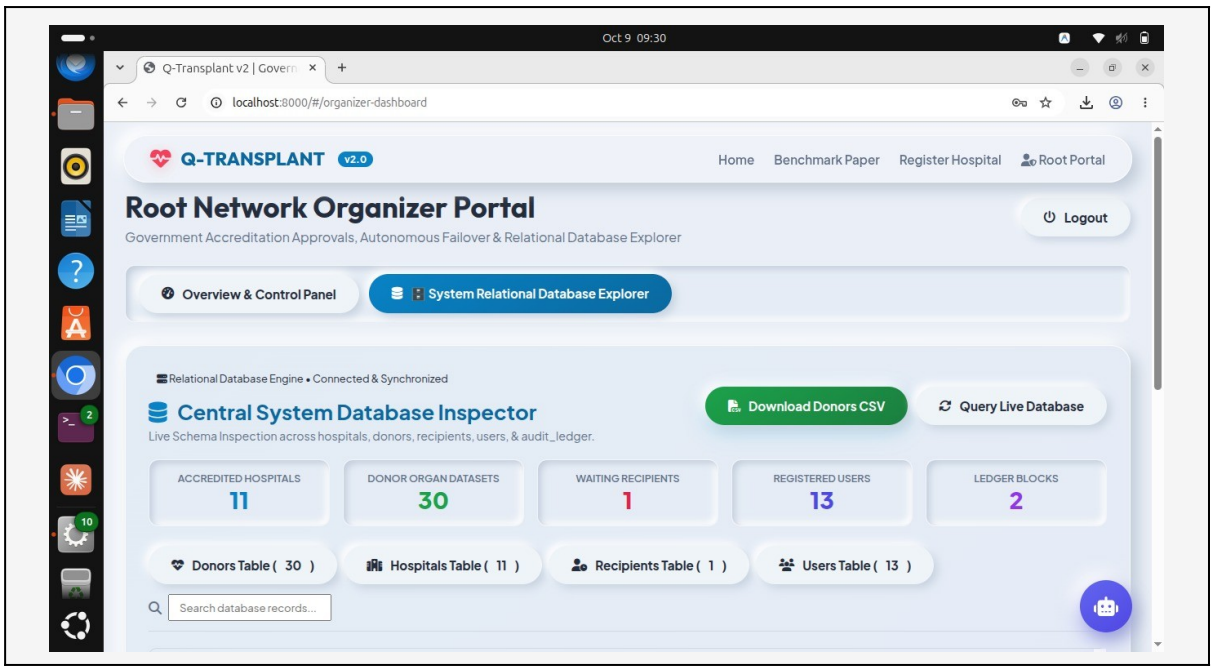


Figure 5.4: Audit log

5.2 TEST PLAN AND TEST CASES

Testing covers implementation-level checks of each module with simulated data. Table 5.1 lists the test cases and the expected results.

Table 5.1: Test cases

ID	Module	Test	Expected result	Result
TC01	Scoring	Evaluate the same donor and recipient twice	Identical score and breakdown	Pass

ID	Module	Test	Expected result	Result
TC02	Scoring	Donor and recipient with incompatible blood groups	Candidate excluded	Pass
TC03	Scoring	Organ type does not match the recipient's need	Candidate excluded	Pass
TC04	Scoring	Recipient flagged as medically ineligible	Candidate excluded	Pass
TC05	Scoring	Two otherwise similar recipients with different urgency	Urgency affects the score as defined by the scoring rule	Pass
TC06	Explainability	Open the score of a match	Per-factor breakdown displayed	Pass
TC07	Privacy	Inspect the hospital-facing response of a match	Score and marker count present; raw alleles absent	Pass
TC08	Access control	User attempts an action outside the role	Request refused	Pass
TC09	State machine	Request an invalid case transition	Rejected; state unchanged	Pass
TC10	Emergency	Raise an emergency case	Emergency state recorded	Pass
TC11	ESP32	Raise an emergency while the device is connected	Device signals the emergency	Pass (backend)
TC12	Audit	Perform a match and a state change	Audit entries created with user and action	Pass
TC13	Search	Run the quantum-inspired search on a simulated pool	Shortlist drawn from the pool	Pass
TC14	Benchmark	Run the harness on the same pool	Comparison with exhaustive search produced	Pass

5.3 TESTING RESULTS

The prototype was evaluated with simulated data only. The project’s automated test suite (pytest) contains 105 tests covering authentication, role-based access control, registration and approval workflows, matching, the restricted HLA view, the case and emergency state machines, device provisioning and the ESP32 WebSocket channel, audit logging and the quantum-inspired search; all 105 tests passed when the suite was run for this report. The 14 test cases of Table 5.1 were also checked by running the matching, state-machine and search modules directly and, for the access-control, privacy, audit and device cases, through the corresponding API tests; all of them passed. TC11 was verified at backend level with a simulated device client that connects to the emergency WebSocket with a provisioned device token and sends acknowledge, processing and donor-found messages; the physical ESP32 hardware (display and buzzer) was not part of the automated run.

The benchmark harness was run on synthetic pools of 50, 200, 1,000 and 5,000 candidates with random scores, using ten pools per size and a fixed search seed of 42, in Python 3.12 on a single process. Table 5.2 summarises the results. Exhaustive search always evaluated every candidate. The quantum-inspired search evaluated between about 1% and 11% fewer candidates, but it returned the same best score in only 5 to 9 of the 10 pools at each size, and its wall-clock time

was higher than that of exhaustive search (about 27 ms against 0.4 ms for 5,000 candidates) because of the sampling overhead. These results show no speed or accuracy advantage for the quantum-inspired search at these pool sizes, and no quantum speedup is claimed.

Table 5.2: Benchmark of exhaustive and quantum-inspired search on simulated pools

Pool size	Exhaustive evaluations	Quantum-inspired evaluations	Mean reduction (%)	Same best score (of 10 pools)	Mean time, exhaustive / quantum-inspired (ms)
50	50	about 45	11.0	6	0.006 / 0.11
200	200	about 185	7.2	9	0.014 / 0.37
1,000	1,000	about 977	2.3	5	0.07 / 3.1
5,000	5,000	about 4,932	1.4	7	0.38 / 27.0

For the deterministic engine, ranking a pool of 1,000 simulated donors for one recipient took a median of about 21 ms, and a pool of 5,000 donors about 116 ms, over ten repeated runs. These timings were measured on a general-purpose machine and are indicative only.

5.4 DISCUSSION

The implemented features show that the platform covers matching, restricted data exposure, case and emergency coordination, audit logging and physical alerting in one prototype. The deterministic score makes each ranking reproducible and explainable, and the restricted view reduces routine exposure of HLA data. The quantum-inspired module is a classical method; its value as a candidate-search aid was judged through repeatable benchmark runs against exhaustive search (Section 5.3), which showed only a small reduction in evaluations and no wall-clock benefit on simulated pools. Because the evaluation used simulated data, the results should not be interpreted as clinical performance.

5.5 ADVANTAGES OF THE SYSTEM

- Explainable and deterministic matching that coordinators can inspect.
- Data minimisation: scores and marker counts instead of raw alleles in routine matching.
- An honest benchmark harness for the quantum-inspired module, with no claim of quantum speedup.
- Role-based access control, validated state machines and audit logs for accountable coordination.
- A physical emergency alert through the ESP32 device.

5.6 LIMITATIONS

- The prototype was evaluated only on simulated data and is not a clinical system.

- The benchmark covers only simulated pools of up to 5,000 candidates with synthetic scores, and it shows no speed or accuracy advantage for the quantum-inspired search.
- The quantum-inspired module runs on classical hardware and provides no quantum speedup.
- The scoring rules are a simplified model and do not replace official allocation policy or laboratory cross-matching [9].
- The platform does not integrate with national registries or hospital information systems.

5.7 CONCLUSION

Q-Transplant is a research prototype of a privacy-aware, explainable decision-support platform for organ transplant coordination. It pairs a deterministic scoring engine with a restricted-access design in which hospitals receive compatibility scores and marker counts instead of raw alleles, adds a quantum-inspired candidate-search module with a benchmark harness for comparison with exhaustive search, and completes the platform with role-based access control, validated case and emergency state machines, audit logging and an ESP32 emergency alert device. Building on graph-based optimisation for kidney exchange [1], [2], the work extends optimisation-oriented allocation into a privacy-aware, real-time coordination architecture. The benchmark showed that the quantum-inspired search gives no practical advantage over exhaustive search at the pool sizes tested, which supports keeping the deterministic ranking authoritative; further work is needed to evaluate the platform beyond simulated data.

5.8 FUTURE ENHANCEMENTS

- **Reproducible benchmarks:** extend the repeatable comparisons of the search module with exhaustive search to larger pools and more realistic score distributions.
- **Stronger privacy techniques:** explore cryptographic or differential-privacy approaches for HLA-derived quantities.
- **Wider optimisation:** combine the scoring engine with kidney-exchange cycle and chain optimisation [1], [2].
- **Integration:** connect the platform with hospital information systems and registries under appropriate governance.
- **Clinical evaluation:** assess the platform with clinicians and regulatory review before any real-world use.
- **Alert device:** extend the ESP32 device with additional signalling options and status feedback.

APPENDIX A - CONFERENCE PAPER CERTIFICATES



Certificate 1 - Aravindh



Certificate 2 - Jovin Roy



Certificate 3 - Priyan

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